MILITARY SPECIFICATION

SHIM, SOLID

GENERAL SPECIFICATION FOR

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for solid shims.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-A-250/8	- Aluminum Alloy 5052, Plate and Sheet
QQ-B-613	- Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip)
QQ-P-416	- Plating, Cadmium (Electrodeposited)
QQ-S-766	- Steel Plates, Sheets, and Strip-Corrosion Resisting
PPP-H-1581	- Hardware (Fasteners and Related Items), Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research, Development and Engineering Center, ATTN: SMCAR-ESC-S, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5365

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MILITARY

MIL-F-495	- Finish, Chemical, Black, For Copper Alloys
MIL-C-5541	- Chemical Conversion Coatings On Aluminum and Aluminum Alloys
MIL-A-8625	- Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-S-70334/1	- Shim, Solid, Round

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME B46.1	- Surface Texture (Surface Roughness, Waviness and Lay)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018).

ASTM

ASTM A109	- Steel, Strip, Carbon, Cold-Rolled
ANSI/ASTM A380	- Cleaning and Descaling Stainless Steel Parts, Equipment, and Systems

(Applications for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103).

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

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2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern (see 6.2).

3.2 Material. Recycled and reclaimed materials shall be used to the maximum extent practicable. Solid shims shall be fabricated of the following materials, as specified (see 3.1).

3.2.1 Aluminum alloy. Aluminum alloy shall be Alloy 5052, Temper H38 (UNS A95052) in accordance with QQ-A-250/8.

3.2.2 Brass. Brass shall be Alloy 260, Temper half hard, (UNS C26000) in accordance with QQ-B-613.

3.2.3 Corrosion resistant steel. Corrosion resistant steel shall be Type 302 (UNS S30200), Temper 1/4 hard, in accordance with QQ-S-766.

3.2.4 Carbon steel. Carbon steel shall be Grade 1010 through Grade 1015 in accordance with ASTM A109, Temper No. 4.

3.3 Protective finish or surface treatment. Solid shims shall be furnished with a protective finish or surface treatment, as specified (see 3.1).

3.3.1 Aluminum alloy. Aluminum alloy shims shall be anodic coated in accordance with MIL-A-8625, Type I, Class 1 or chemically treated in accordance with MIL-C-5541, Class 1A, as specified (see 3.1).

3.3.2 Brass. Brass shims shall have a black chemical finish in accordance with MIL-F-495.

3.3.3 Carbon steel. Carbon steel shims shall be cadmium plated in accordance with QQ-P-416, Type II, Class 3.

3.3.4 Corrosion resistant steel. Corrosion resistant steel shims shall be cleaned, descaled and passivated in accordance with ANSI/ASTM A380.

3.4 Dimensions and tolerances. Dimensions and tolerances shall be in accordance with the applicable sheet (see 3.1) and as specified herein and shall apply after protective finish, if applicable.

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3.5 Surface roughness. Shims shall have a maximum surface roughness of 63 microinches, determined in accordance with ANSI/ASME B46.1.

3.6 Cracks. Shims shall be free from cracks, flaws and pits in any location. A crack is a clean crystalline fracture passing through or across the grain boundaries without inclusion of foreign elements.

3.7 Workmanship. Shims shall be free from surface contamination, tool marks and other imperfections which may adversely affect useability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Quality conformance inspection. Quality conformance inspection shall be as specified in Table I.

TABLE I. QUALITY CONFORMANCE INSPECTION

INSPECTION	REQUIREMENT PARAGRAPH	TEST PARAGRAPH
Protective finish or surface treatment (as applicable)	3.3	4.3.2, 4.3.3
Dimensions	3.4	4.3.1
Surface roughness	3.5	4.3.1
Workmanship	3.7	4.3.1

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4.2.1 Inspection lot. An inspection lot shall consist of all shims covered by a single specification sheet, of the same material and finish, nominal size, produced by the same manufacturer under essentially the same conditions, and submitted for acceptance at one time.

4.2.2 Rejected lots. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units and resubmit for inspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots and shall be clearly identified as reinspected lots.

4.2.3 Sampling for visual and dimensional examination. A random sample of shims shall be taken from each lot in accordance with MIL-STD-105 Inspection Level II. The Acceptable Quality Level (AQL) shall be as indicated in Table II.

4.2.4 Inspection of packaging. The sampling and inspection of the preservation, packing and container marking shall be in accordance with the requirements of PPP-H-1581.

4.3 Methods of inspection.

4.3.1 Visual and dimensional examination. Samples of shims shall be examined to verify conformance with this specification and applicable specification sheet. Examination shall be conducted in accordance with Table II.

4.3.1.1 Examination for evidence of cracks, flaws or pits. Samples shall be examined for evidence of cracks, flaws or pits using a 3X or 5X magnification visual aid.

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TABLE II. CLASSIFICATION OF DEFECTS

CATEGORY	DEFECT	INSPECTION METHOD
CRITICAL	NONE DEFINED	
Major	AQL = 2.5 percent defective	
101	Inside diameter, not as specified (see 3.4)	CIE 1/
102	Outside diameter, not as specified (see 3.4)	CIE 1/
103	Thickness, not as specified (see 3.4)	CIE 1/
104	Evidence of cracks, flaws or pits (see 3.6)	Visual
Minor	AQL = 4.0 percent defective	
201	Protective finish or surface treatment missing or incomplete (see 3.3) (as applicable)	Visual
202	Surface roughness, not as specified (see 3.5)	CIE 1/
203	Workmanship (see 3.7)	Visual

1/ Commercial inspection equipment

4.3.2 Protective finish. Samples of aluminum alloy, brass and carbon steel shims shall be inspected for adequacy of plating in accordance with 3.3.1, 3.3.2 and 3.3.3.

4.3.3 Surface treatment. Samples of corrosion resistant steel shims shall conform to the inspection after cleaning tests specified in ANSI/ASTM A380.

4.3.4 Material inspection. Material inspection shall consist of certification supported by verifying data that the materials used in fabricating the shims are in accordance with the applicable paragraphs listed in 3.2 prior to such fabrication.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with PPP-H-1581 (see 6.2).

6. NOTES

6.1 Intended use. Shims acquired under this specification are intended for maintaining a predetermined distance between two surfaces.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification and applicable specification sheet.

b. Applicable specification sheet part number (see 3.1).

c. Level (degree) of protection in accordance with PPP-H-1581, ordering data (see 5.1).

6.3 Keyword listing.

Shim
Washer, Flat
Spacer, Ring

Custodians:
Army - AR
Air Force - 99

Review activities:
Army - AV, MI
DLA - IS

User activity:
NSA - NS

Preparing activity:
Army - AR

Agent:
DLA - IS

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